成鱼

作业纸

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10.20 设第一台卡诺热机从高温热库吸收的热量和向低温热库放出的热量为 $Q_1, Q_2, Q_3 Q_4$

则 $\eta_1 = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$

$\eta_2 = 1 - \frac{Q_4}{Q_3}$

又 $Q_3 = Q_2$ 则 $\eta_2 = 1 - \frac{Q_4}{Q_2}$

$\eta = \frac{Q_1 - Q_4}{Q_1} = 1 - \frac{Q_2}{Q_1} + \frac{Q_2 - Q_4}{Q_1} = \eta_1 + \frac{Q_3 - Q_4}{Q_1}$

$= \eta_1 + \frac{Q_2}{Q_1} \cdot \left(1 - \frac{Q_4}{Q_2}\right)$

$= \eta_1 + (1-\eta_1)\eta_2$ 证毕

$\eta = 1 - \frac{T_2}{T_1} + \left[1 - \left(1 - \frac{T_2}{T_1}\right)\right]\left(1 - \frac{T_3}{T_2}\right)$

$= 1 - \frac{T_2}{T_1} + \frac{T_2}{T_1} - \frac{T_2}{T_1} \cdot \frac{T_3}{T_2}$

$= 1 - \frac{T_3}{T_1}$ 证毕

10.24 煤燃烧发热量 $Q_1 = 1 \times 3.34 \times 10^7 J = 3.34 \times 10^7 J$

$T_1 = 483K$，$T_2 = 333K$，$T_3 = 288K$

$\frac{Q_1}{|Q_2|} = \frac{T_1}{T_2}$　　$A = Q_1 - Q_2 = \frac{T_1 - T_2}{T_1} Q_1$

解 $Q_2 = \frac{T_2}{T_1} Q_1$

$Q_2' = A + Q_3$

$\frac{Q_3}{Q_2' - Q_3} = \frac{T_3}{T_2 - T_3}$

解 $Q_2' = \frac{T_2}{T_2 - T_3} A = \frac{T_2(T_1 - T_2)}{T_1(T_2 - T_3)} Q_1$

$Q = Q_2 + Q_2' = \frac{T_2(T_1 - T_3)}{T_1(T_2 - T_3)} Q_1$

$= \frac{333 \times (483 - 333)}{483 \times (333 - 288)} \times 3.34 \times 10^7 J$

$= 9.98 \times 10^7 J$

$n = \frac{Q}{Q_1} = 2.99$

即 水得到的热量是煤所发热量的 2.99 倍

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11.1 气体对外做功 $A=\int PdV$

$$=\frac{1}{2}\times(6+8)\times10^5\times10^{-3}+\frac{1}{2}\times(8+4)\times10^5\times10^{-3}$$

$$=1.3\times10^3\,J$$

气体内能的变化量 $\Delta E=\frac{i}{2}(P_cV_c-P_aV_a)$

$$=\frac{5}{2}\times(4\times10^5\times3\times10^{-3}-6\times10^5\times10^{-3})=1.5\times10^3\,J$$

气体吸的热 $Q=\Delta E+A=2.8\times10^3\,J$

熵变 $\Delta S=C_v\ln\frac{T_b}{T_a}+R\ln\frac{V_c}{V_a}=\frac{5}{2}\times8.31\times\ln\frac{4\times3}{6\times1}+8.31\times\ln\frac{3}{1}$

$$=23.5\ J/K$$

11.2 $\Delta S=\int_{T_1}^{T_2}\frac{dQ}{T}$

又 $Q_1=C_1mT$

$dQ_1=C_1mdT$

同理 $dQ_2=C_2mdT$

-10℃的冰变为0℃的冰过程中

$\Delta S_1=\int_{T_1}^{T_2}\frac{dQ}{T}=\int_{T_1}^{T_2}\frac{C_1mdT}{T}=C_1m\ln\frac{T_2}{T_1}$

0℃的冰变为0℃的水过程中

$\Delta S_2=\frac{\lambda m}{T_2}$

0℃的水变成100℃的水过程中

$\Delta S_3=C_2m\ln\frac{T_3}{T_2}$

100℃的水变成水蒸气过程中

$\Delta S_4=\frac{Lm}{T_3}$

故总熵变 $\Delta S=\Delta S_1+\Delta S_2+\Delta S_3+\Delta S_4=30\times2.1\times\ln\frac{273}{243}+30\times4.2\times\ln\frac{373}{273}+30\times\frac{334}{273}+30\times\frac{2260}{373}$

$$=268\ J/K$$

11.5 $V_s=\frac{\Delta S}{\Delta t}=\frac{\Delta Q}{\Delta T\cdot\Delta t}$

$$=\frac{P\Delta t}{\Delta T\cdot\Delta t}$$

$$=\frac{P}{\Delta T}=\frac{20\times10^3}{285}\ J/(K\cdot s)=70\ J/(K\cdot s)$$

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$\Delta S=\frac{\Delta Q}{\Delta T}$

若水的机械能全部转换成内能

$mgh=cm\Delta T$

$\Delta T=T_2-T_1$ 且 $T_1=285K$

$m=Q\Delta t$

$\Delta S=\int_{T_1}^{T_2}\frac{dQ}{T}=\int_{T_1}^{T_2}\frac{cmdT}{T}=cm\ln\frac{T_2}{T_1}=cm\ln\frac{T_1+\frac{gh}{c}}{T_1}$

又 $\frac{gh}{c}\ll T_1$ 则 $\Delta S=cm\cdot\frac{gh}{cT_1}=\frac{mgh}{T_1}$

$$=\frac{1500\times10^3\times9.8\times74}{285}=3.8\times10^6\ J/K$$

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11.7 (1) $\Delta S_1 = \int_{T_1}^{T_2} \frac{dQ}{T} = \int_{T_1}^{T_2} \frac{cmdT}{T} = cm\ln\frac{T_2}{T_1}$

$\Delta S_2 = \frac{Q}{T_2} = \frac{-cm(T_2-T_1)}{T_2}$

$\Delta S = \Delta S_1 + \Delta S_2$

解 $\Delta S = 184\ J/K > 0$ 即系统的总熵增加

(2) 将水放置50℃的恒温热库达到平衡

$\Delta S_{11} = cm\ln\frac{T_3}{T_1}$

$\Delta S_{12} = \frac{-cm(T_3-T_1)}{T_3}$

再移到100℃的恒温热库达到平衡

$\Delta S_{21} = cm\ln\frac{T_2}{T_3}$

$\Delta S_{22} = \frac{-cm(T_2-T_3)}{T_2}$

$\Delta S = \Delta S_{11} + \Delta S_{12} + \Delta S_{21} + \Delta S_{22}$

$= cm\ln\frac{T_2}{T_1} - \frac{cm(T_3-T_1)}{T_3} - \frac{cm(T_2-T_3)}{T_2}$

$= 97\ J/K > 0$ 即系统的总熵增加，但比(1)增加得少

11.8 (1) 冰水的熵变

$\Delta S_1 = \frac{-\lambda m}{T} = \frac{-334\times10^3\times0.9}{273}\ J/K = -1.1\times10^3\ J/K$

环境的熵变

$\Delta S_2 = \frac{\lambda m}{T} = 1.1\times10^3\ J/K$

总熵变 $\Delta S = \Delta S_1 + \Delta S_2 = 0$

(2) 冰水的熵变

$\Delta S_1' = -\Delta S_1 = 1.1\times10^3\ J/K$

恒温箱的熵变

$\Delta S_2' = \frac{-\lambda m}{T'} = \frac{-334\times10^3\times0.9}{373}\ J/K = -0.81\times10^3\ J/K$

总熵变 $\Delta S' = \Delta S_1' + \Delta S_2'$

$= 290\ J/K$

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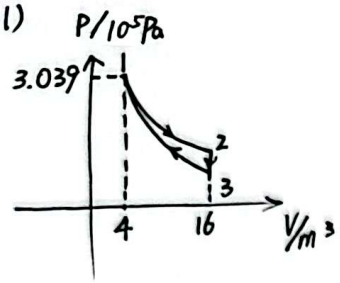
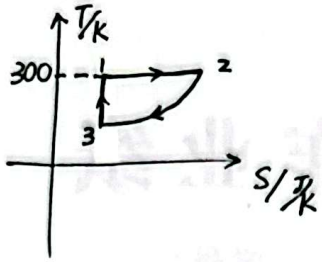
11.9 (1)





(2) 1→2 过程

气体做的功

$$A_{12} = \nu R T_1 \ln\frac{V_2}{V_1} = P_1V_1 \ln\frac{V_2}{V_1} = 3.039\times10^5\times4\times\ln\frac{16}{4}$$

$$= 1.69\times10^6 J$$

熵变

$$\Delta S_{12} = \frac{Q}{T_1} = \frac{A_{12}-\Delta E}{T_1}$$

$$\Delta E = 0$$

解 $\Delta S_{12} = \frac{A_{12}}{T_1} = \nu R \ln\frac{V_2}{V_1} = 5.63\times10^3 J/K$

2→3 过程

$A_{23} = 0$ (体积不变)

熵变

$$\Delta S_{23} = \int_{T_1}^{T_2}\frac{dQ}{T} = \int_{T_1}^{T_2}\frac{\nu C_{V,m}dT}{T} = \nu C_{V,m}\ln\frac{T_2}{T_1}$$

又 $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$ 解 $\Delta S_{23} = \nu C_{V,m}(\gamma-1)\ln\frac{V_1}{V_2}$

$$= -5.63\times10^3 J/K$$

3→1 过程

$$A_{31} = \Delta E = \frac{i}{2}(P_2V_2 - P_1V_1)$$

$$= -1.30\times10^6 J$$

熵变

$$\Delta S_{31} = 0$$

气体对外做总功 $A = A_{12}+A_{23}+A_{31} = 3.9\times10^5 J$

熵变 $\Delta S = S_{12}+S_{23}+S_{31} = 0$

11.10 易知混合平衡态的温度 $T = \frac{T_1+T_2}{2}$

低温液体的熵变

$$\Delta S_1 = \int_{T_1}^{T}\frac{\frac{m}{M}C_p dT}{T} = \frac{m}{M}C_p\ln\frac{T}{T_1}$$

同理高温液体的熵变

$$\Delta S_2 = \int_{T_2}^{T}\frac{\frac{m}{M}C_p dT}{T} = \frac{m}{M}C_p\ln\frac{T}{T_2}$$

$(T_1+T_2)^2 > 4T_1T_2$

总熵变 $\Delta S = \Delta S_1+\Delta S_2 = \frac{m}{M}C_p\ln\frac{T^2}{T_1T_2} = \frac{m}{M}C_p\ln\frac{(T_1+T_2)^2}{4T_1T_2} > 0$ (M为液体的摩尔质量)

故熵增加

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